

# PROJECT PRESENTATION

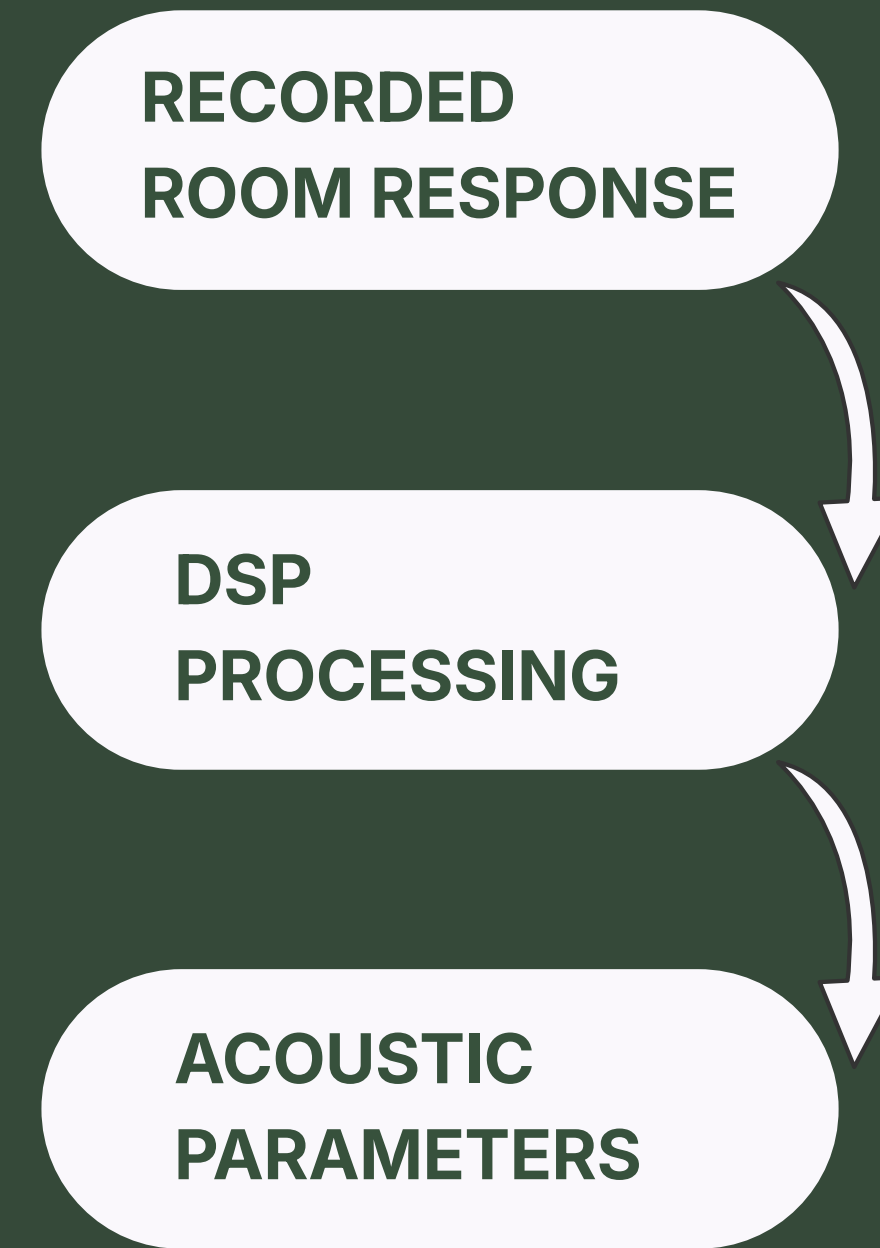
BY 2305040,2305051,2305053

# RESONA

RESONA is a web-based audio analyzer which works on the basis of

a single audio clip where a sudden noise like a clap or thump is present.

This acts as an impulse response for the space and we can derive different calculations based on this response.



# RESONA FEATURES

For a specific type of room, RESONA calculates different acoustic parameters which can help with building a space with the required auditory demands.



## RT60

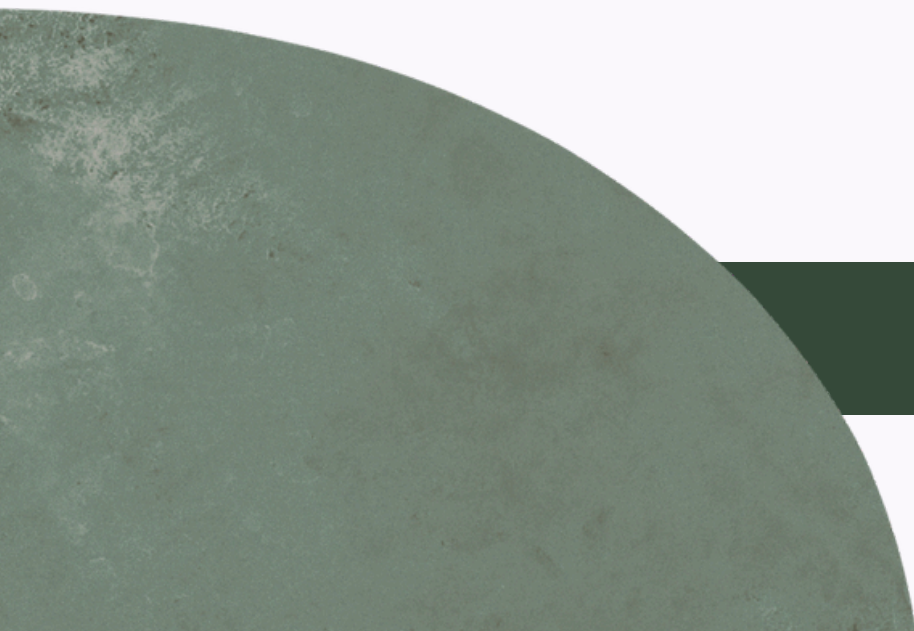
RT60 is the time (in seconds) it takes for sound energy in an enclosed space to decrease by 60 decibels (dB) after the sound source stops

## ROOM MODES

Room modes are the standing waves that form at particular frequencies based on the dimensions of the room.

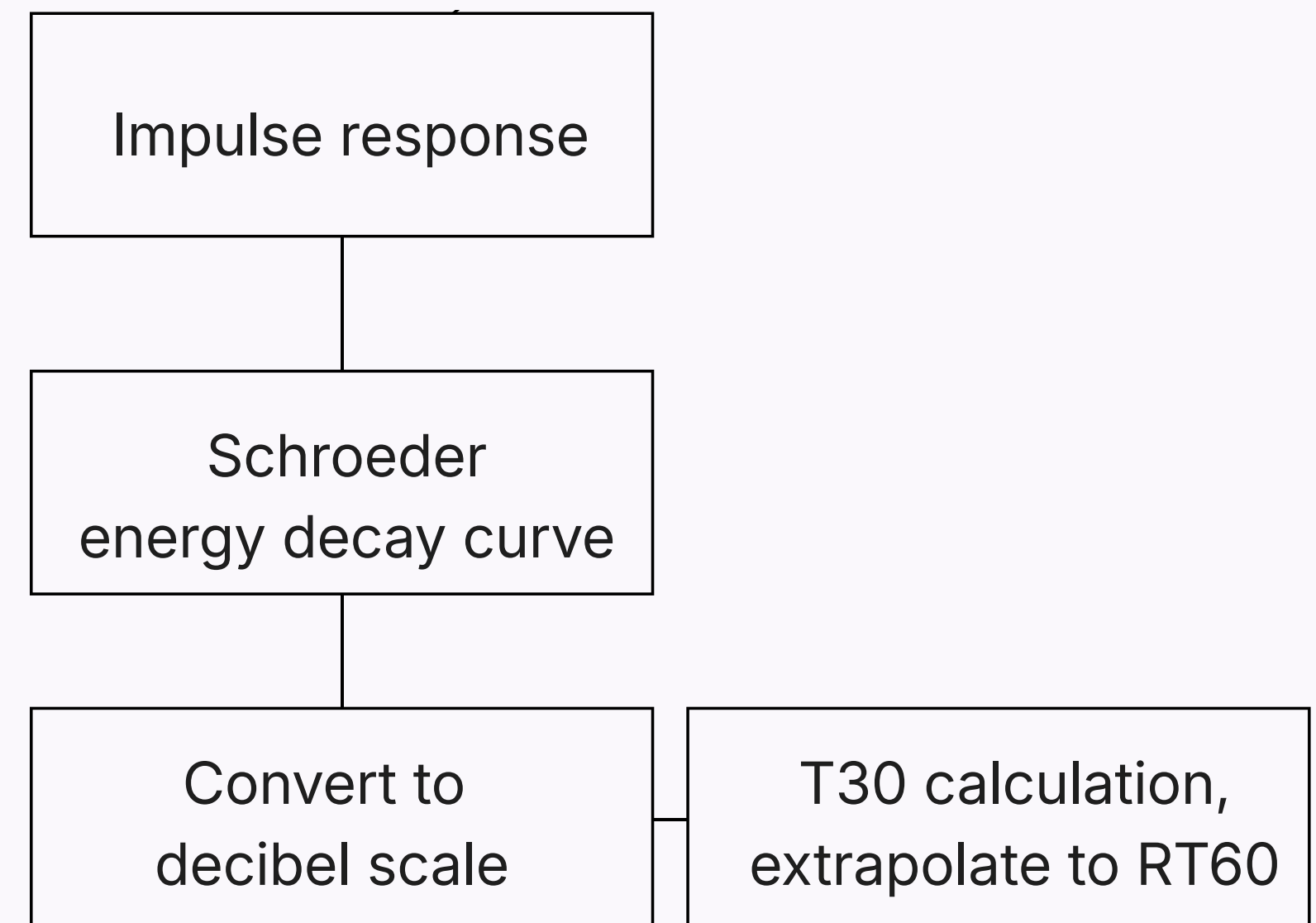
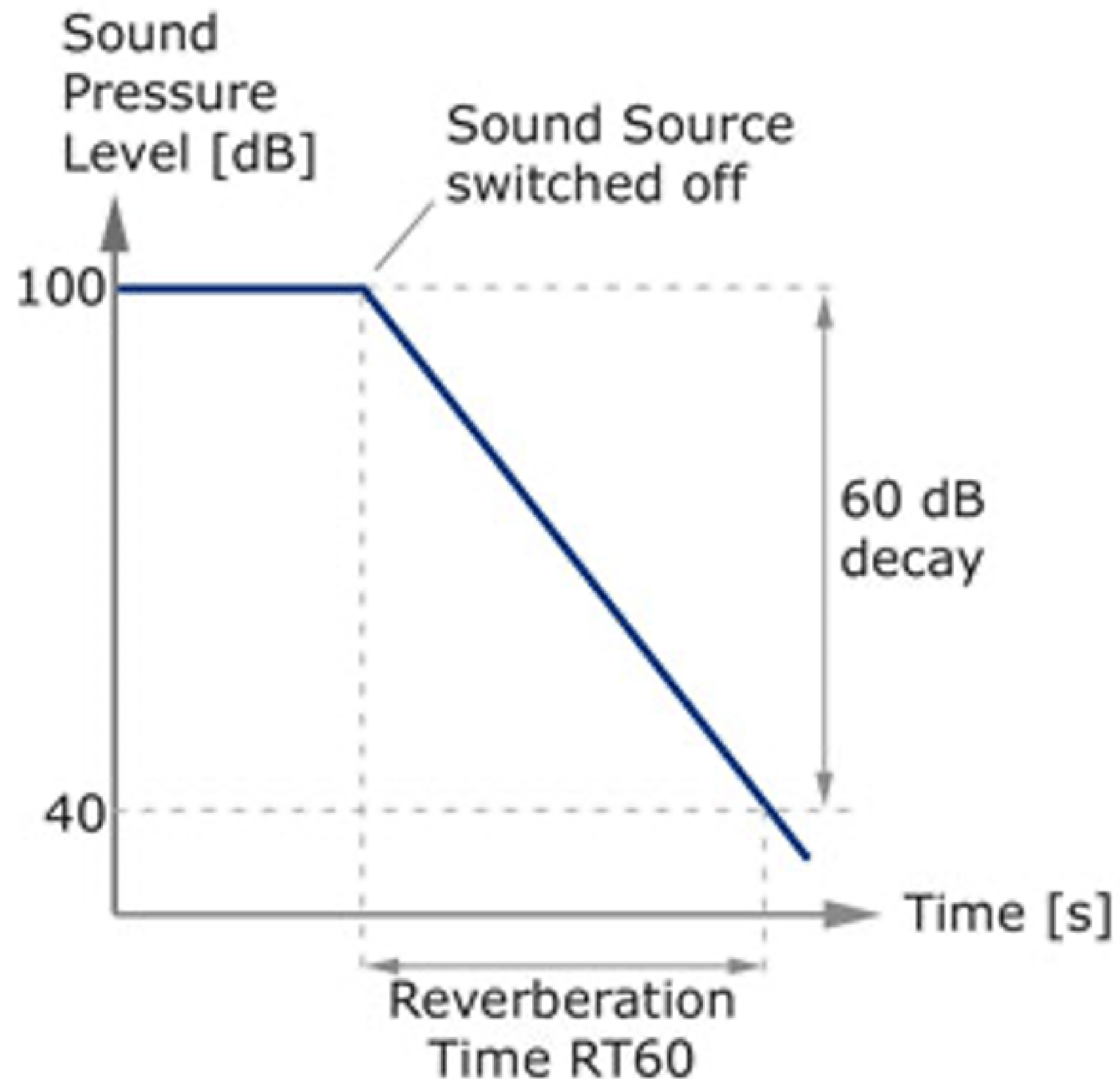
## 3D WATERFALL

3D waterfall displays how a specific clique of frequencies in the signal decay over time and space.



# RT60

RT60 measures how long it would take for the sound energy to decrease by 60 dB after the sound source stops.



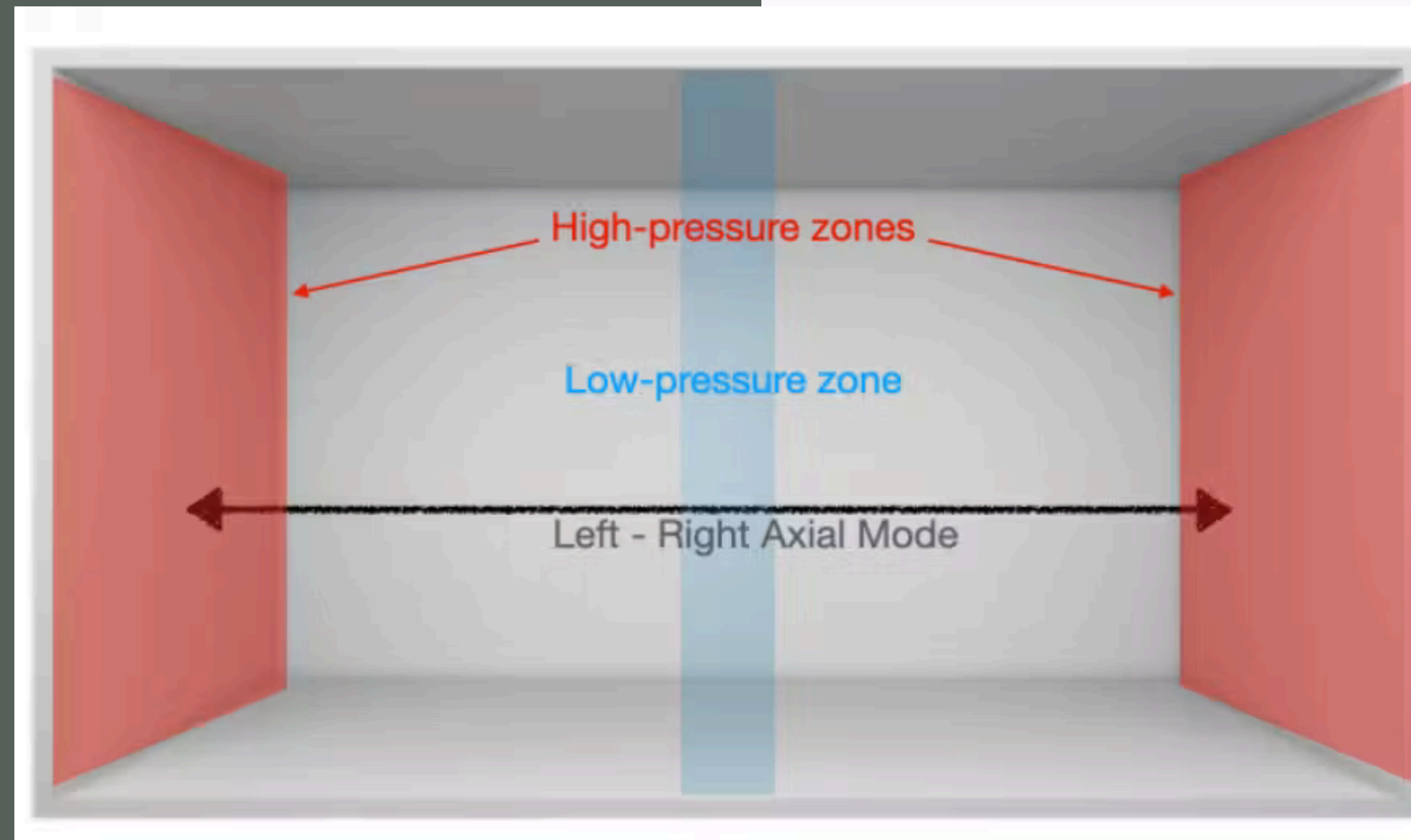
## FEATURE DESCRIPTION

# ROOM MODES

Sound waves reflect between walls. At certain frequencies, the reflected waves line up with each other and create standing waves. These resonant frequencies are called:

Room Modes.

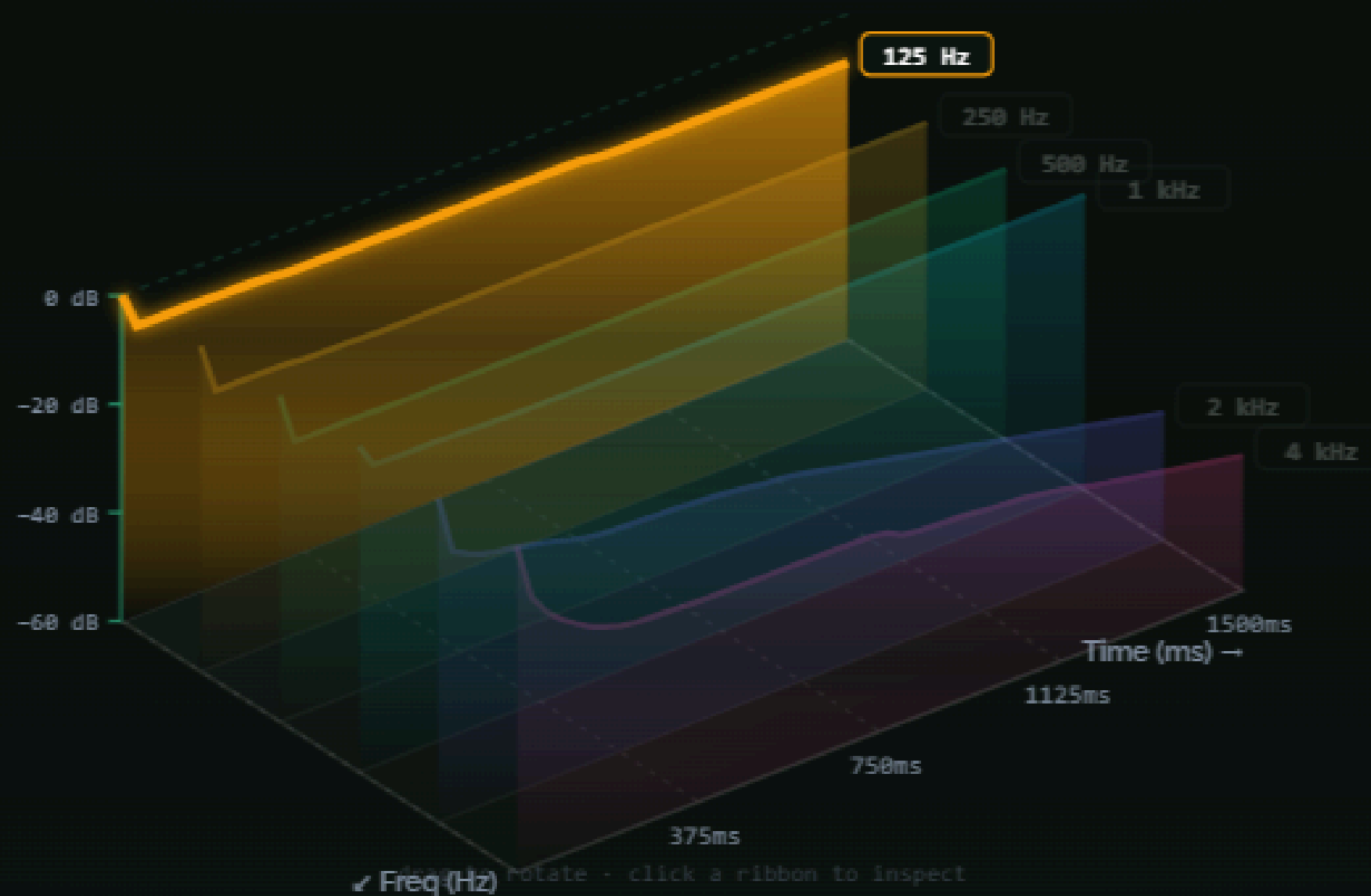
The room dimensions directly determine its resonant frequencies. The room modes can cause some frequencies to become unnaturally amplified or diminished at certain points.

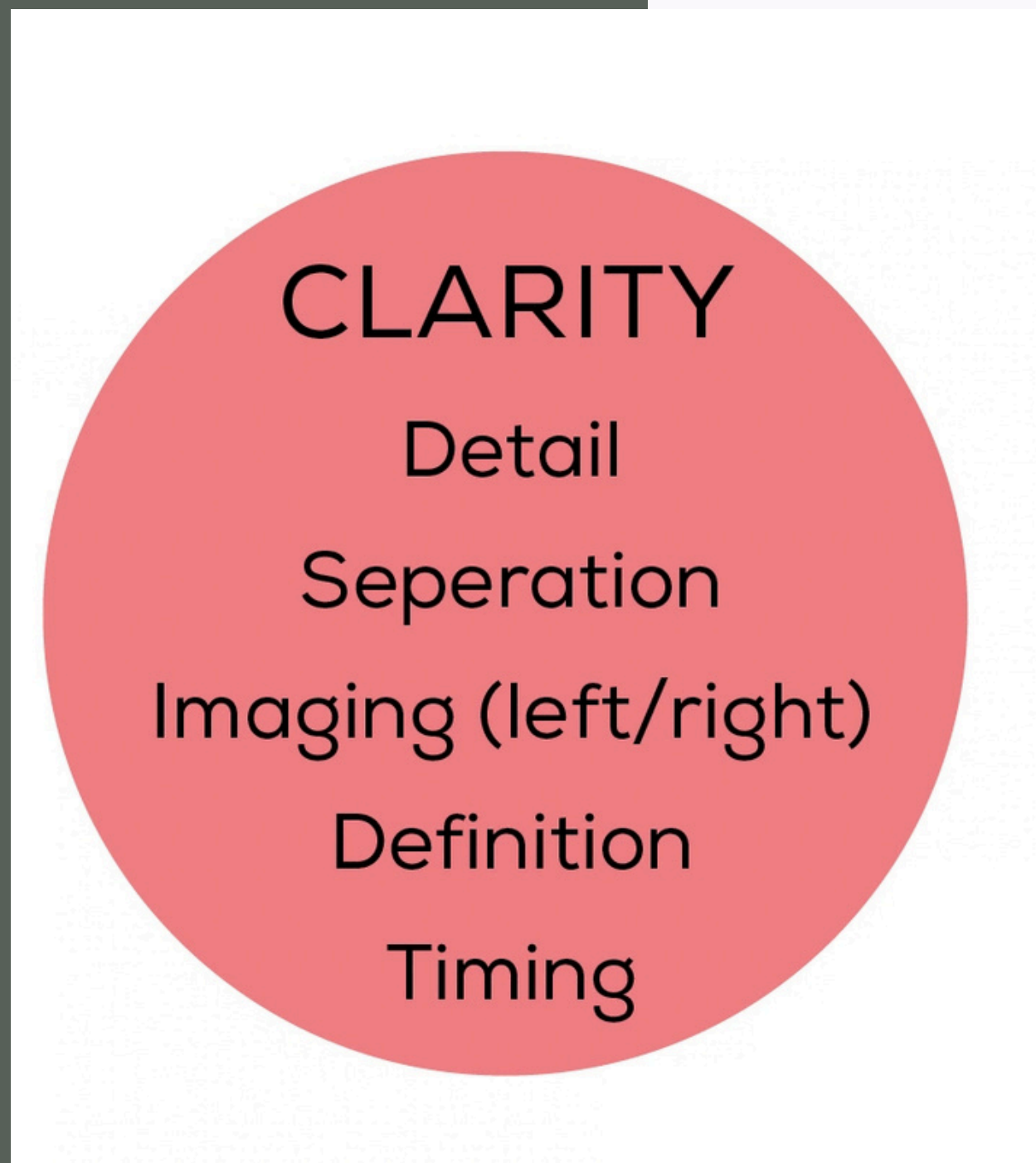


## FEATURE DESCRIPTION

# WATERFALL PLOT

A waterfall plot shows how the frequency spectrum changes over time, allowing us to identify frequencies that persist or 'ring' longer than others. The signal is divided into small time windows. For each window, we calculate an FFT.





## FEATURE DESCRPTION

# CLARITY & CONCISENESS

For any room, we know how big of a role audio clarity plays as it disperses the vocal message throughout the audience.

To better help understand this key parameter, we have kept a metric for the clarity factor.

Clarity measures the relationship between early sound energy and late sound energy.

C50 is for speech clarity whereas C80 is for music clarity.

# RESONA WORKFLOW



01

RECORD



02

DSP  
PROCESSING



03

ACOUSTIC  
PARAMETERS



04

RECOMMENDATIONS  
FOR MATERIAL



# THANK YOU!

VISIT RESONA AT: [HTTPS://RESONA-RT60-ROOM-ACOUSTICS-ANALYZER.VERCEL.APP/](https://resona-rt60-room-acoustics-analyzer.vercel.app/)

